

GRB 090719

AGENTIC AI REPORT BY VIKAS CHAND, KHUSHBOO SHARMA, AND JAGDISH C. JOSHI

ABSTRACT

We present the complete timing and time-resolved spectral analysis of the single-pulse burst GRB 090719 (*Fermi*/GBM trigger bn090719063), the sixth burst of a 106-burst single-pulse campaign run on a staged, gate-reviewed pipeline. The burst is compact and bright: $T_{90} = 13.39 \pm 0.22$ s, with the campaign’s fastest canonical variability — the windowed estimator finds an 8.9 ± 1.9 ms detection ($\Delta t = 0.4$ s segment; rung significance $z = 2.8$) — while the wavelet method lands on a new grid rung (256 ± 20 ms, after four bursts on 215 ms) and the Haar cross-check gives 679 ± 43 ms: the estimator spread now spans a factor of 76 within one burst. The pulse is again *mixed* rather than FRED-like ($\varphi = 0.408 \pm 0.023$, $R^2 = 0.96$) — the second consecutive non-FRED, at nearly the same asymmetry as burst #5. The window-scanned lag is $\tau = 0.80^{+0.03}_{-0.02} \pm 0.08_{\text{win}}$ s at 93σ , the campaign’s most significant. Ten-block spectroscopy shows the cleanest hard-to-soft evolution so far: in the uniform Band convention E_p falls near-monotonically from 624 keV to a bound-railed ≤ 30 keV (one block-3 excursion) while α runs from -0.14 through a $+0.08$ maximum to -1.16 , crossing the synchrotron line of death mid-burst — the third consecutive burst in which the violation is an epoch property. The integrated spectrum is won by the free-scale smoothly-broken power law ($E_b = 188^{+30}_{-18}$ keV, $\beta = -2.77$), in a statistical tie with a BB-bearing composite at $\Delta\text{AIC} = 0.2$. Three blocks prefer blackbody-bearing models, all as statistical ties ($\Delta\text{AIC} \leq 1.6$), all edge-clean; none rises to a robust thermal claim. The integrated winner’s effective-area factors sit off-rail for the first time in the campaign ($n_8 = 1.01$, $b_1 = 1.14$), though block-level winners still rail b_1 in six of ten blocks. The 24×11 grid renders under the no-model-dropped rule with two labeled engine-refusal cells; the last block’s raw winner is bound-railed and disclosed invalid.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

GRB 090719 triggered the *Fermi* Gamma-ray Burst Monitor (Meehan et al. 2009) on 2009 July 19. As burst #6 of the single-pulse campaign it runs on the pipeline frozen from bursts #1–#2. It adds three data points to the accumulating cross-burst picture: a second consecutive mixed-shape pulse at $\varphi \approx 0.40$ (suggesting the non-FRED class is common, not exceptional); a CWT value on a *new* grid rung, sharpening the campaign’s characterization of the wavelet estimator’s discrete scale ladder; and a third consecutive epoch-resolved crossing of the synchrotron line of death.

2. OBSERVATIONS, DETECTORS, AND APPROVED SELECTIONS

We use TTE data from NaI detectors n_7 , n_8 and BGO b_1 (Figure 1) — the joint-smallest approved detector set of the campaign (with burst #1) — with the spectral convention of Chand et al. (2020): NaI 8–900 keV excluding the 30–40 keV K-edge, BGO 0.2–38 MeV. Back-

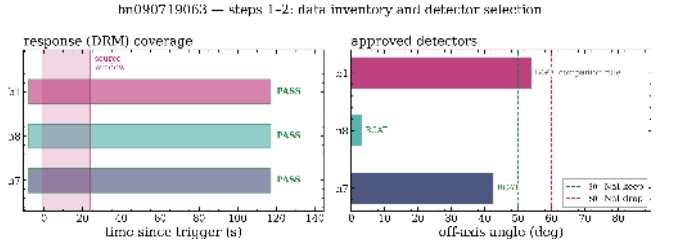


Figure 1. Detector selection for GRB 090719: source angles with the 50° rule; selected detectors n_7 , n_8 , b_1 .

ground and source selections are the human-approved Stage-1 decisions (source window $(-0.91, 24.00)$ s; Figures 2–3). Effective-area factors float in $[0.8, 1.2]$. The integrated winner carries $n_8 = 1.01$ and $b_1 = 1.14$ — the campaign’s first off-rail integrated fit — but block-level winners still rail b_1 (at 1.2 in blocks 2, 5, 6, 8; at 0.8 in blocks 7 and 9), so the disclosure caveat travels with the time-resolved statements.

3. TIMING ANALYSIS

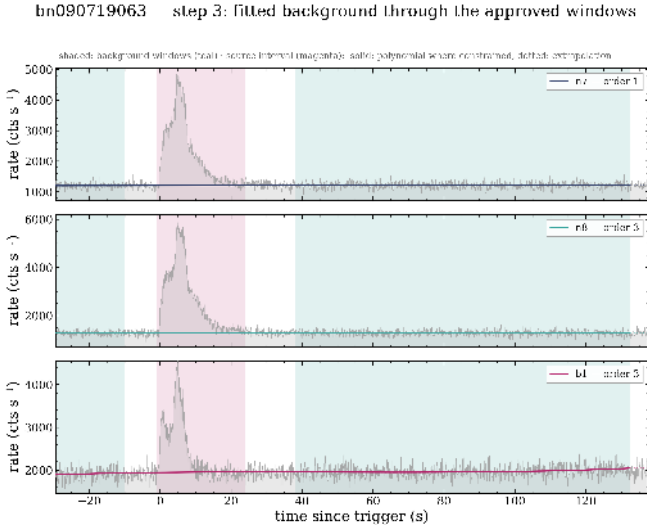


Figure 2. Approved background intervals and per-detector polynomial fits.

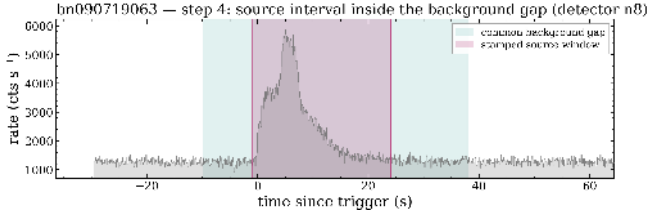


Figure 3. Approved source interval over the background-subtracted light curve.

All timing runs on the approved selections, reference NaI n_8 .

3.1. Durations

The windowed duration is $T_{90} = 13.39 \pm 0.22$ s spanning $[0.94, 14.34]$ s (covariance-true Monte Carlo; window not truncated), with $T_{50} = 4.73$ s ($T_{50}/T_{90} = 0.35$).

3.2. Pulse Shape

The Kocevski et al. (2003) profile wins the shape comparison, and the two-sigmoid of Gowri et al. (2025) gives $\varphi = s_l/s_r = 0.408 \pm 0.023$ ($R^2 = 0.96$): *mixed*, the second consecutive non-FRED classification, and statistically indistinguishable from burst #5’s 0.40 ± 0.05 (Figure 4). Two of the campaign’s five analysed pulses now sit in a narrow mixed band near $\varphi = 0.4$ — early evidence for a shape class, to be tested as the sample grows.

3.3. Minimum Variability Timescale

All three estimators return significant measurements (Figure 5), and the spread is the campaign’s widest: the windowed framework of Bala et al. (2026) — the canonical value — finds 8.9 ± 1.9 ms ($\Delta t = 0.4$ s segment, detection; the selected rung’s significance is $z = 2.8$, below

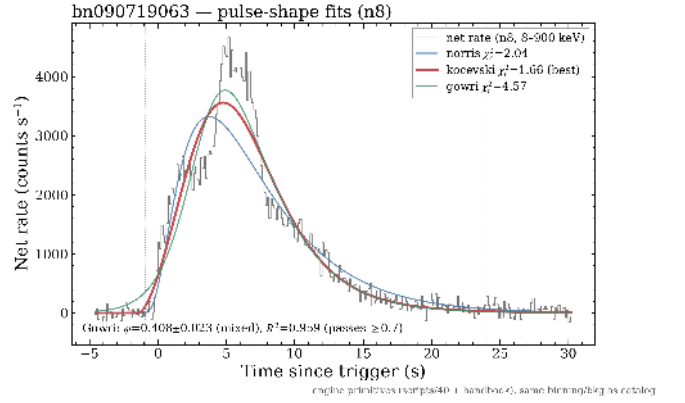


Figure 4. Pulse-shape fits to the net light curve; Kocevski wins; $\varphi = 0.408 \pm 0.023$, the second consecutive mixed classification.

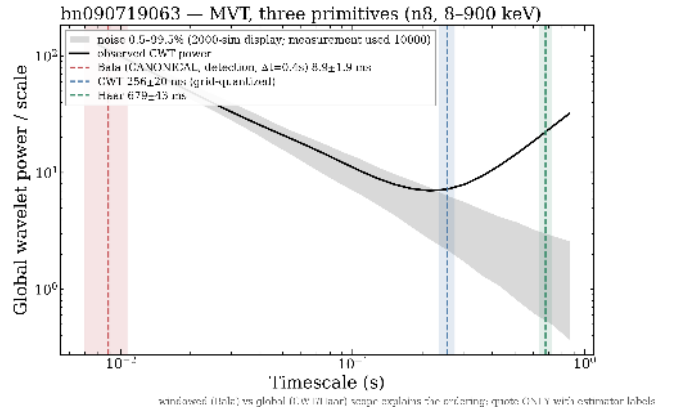


Figure 5. Minimum-variability estimators with explicit labels: canonical windowed 8.9 ± 1.9 ms; CWT 256 ± 20 ms (new grid rung); Haar 679 ± 43 ms.

the campaign’s 3σ comfort line, so the value carries that caveat); the global CWT (Vianello et al. 2018) returns 256 ± 20 ms; the Haar cross-check (Golkhou & Butler 2014) gives 679 ± 43 ms. The CWT value breaks the four-burst run on the 215 ms rung, landing on the adjacent rung of the discrete wavelet-scale ladder — consistent with the campaign’s standing finding that CWT values are quantized by the estimator’s grid, and now showing the ladder itself (215, 256 ms are neighboring scales). The factor-76 spread between canonical and Haar within a single burst is the strongest single-burst demonstration yet that “the MVT” is estimator-defined.

3.4. Spectral Lag

Scanning the cross-correlation fit window around the peak (half-widths 1.6–4.2 s; Figure 6) gives $\tau = 0.80^{+0.03}_{-0.02} \pm 0.08_{\text{win}}$ s (25–50 keV vs. 100–300 keV, soft lagging hard; detector n_8 , 16 ms resolution, peak significance 93σ — the campaign’s most significant lag). Scan members run 0.77–0.93 s; the quoted value is the

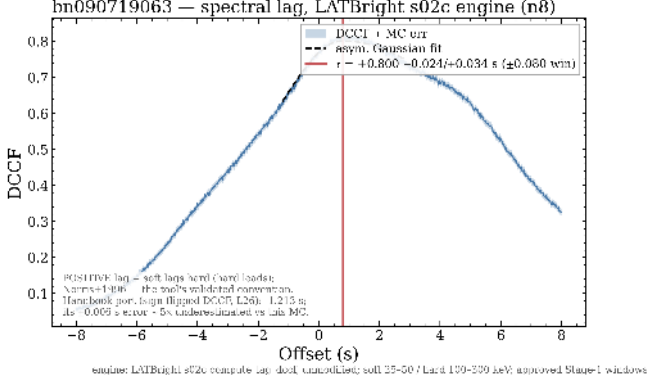


Figure 6. Cross-correlation lag with the window-scan systematic; the quoted value is the pulse-scaled member closest to the scan median.

pulse-scaled member closest to the scan median. The lag fraction $\tau/T_{90} \approx 6.0\%$ extends the campaign’s non-constant lag-width sequence (8.5%, 5.4%, 2.6% so far).

3.5. Per-band Durations

Per-band light curves and the $T_{90}(E)$ relation are shown in Figure 7; the population slope of Qin et al. (2013) is drawn for reference only.

4. SPECTRAL BINNING

Ten Bayesian blocks plus the integrated spectrum (Figure 8), from the frozen 3ML Bayesian-block + significance hybrid (blocks computed 2026-08-12, reused unchanged).

5. TIME-RESOLVED AND TIME-INTEGRATED SPECTROSCOPY

5.1. Method

All 24 campaign models fit to every bin (PGstat; AIC ranking (Akaike 1974)); no LLE/LAT data exist for this burst (inclusion is gated on data availability, never detection significance). The 24×11 grid renders under the no-model-dropped rule with two labeled engine-refusal cells (BandR+CPL, block 4; DSBPLfree, block 9); per-bin all-model parameter tables accompany the products.

5.2. Integrated Spectrum

The integrated spectrum $(-0.14, 20.71)$ s is won by the free-scale SBPL (Figure 9): $\alpha = -0.887^{+0.038}_{-0.030}$, $E_b = 188^{+30}_{-18}$ keV, $\beta = -2.77^{+0.17}_{-0.27}$, sharpness scale 0.32 ± 0.07 (AIC = 5007.7) — but the runner-up, the BB-bearing composite SBPL+BB+CPL, sits at $\Delta\text{AIC} = 0.2$: a statistical tie between a pure one-break shape and a thermal-loaded alternative, which we report as unresolved. In the uniform Band convention the integrated $\alpha = -0.705$, essentially on the synchrotron line of death

— fitting, for a burst whose time-resolved track crosses that line mid-burst.

5.3. Time-Resolved Story: the Cleanest Hard-to-Soft Track

In the uniform Band convention E_p falls near-monotonically across the ten blocks — $624 \rightarrow 394 \rightarrow 202 \rightarrow 319 \rightarrow 237 \rightarrow 179 \rightarrow 119 \rightarrow 75 \rightarrow 54 \rightarrow 30$ keV, with a single excursion at block 3 — spanning a factor ≥ 21 (the 30 keV endpoint is the engine’s E_p floor and that fit is bound-railed invalid, so the true endpoint is at or below it), while α runs from -0.14 through a $+0.08$ maximum to -1.16 (Figure 10). The line of death is approached from above in the first three blocks ($\alpha = -0.14, -0.05, +0.08$), crossed near block 3, and left far behind by the decay — the third consecutive burst showing the violation as an epoch property, here with the smoothest parameter track. Model preference is fragmented with thin margins throughout (all eleven spectra are decided by $\Delta\text{AIC} < 2$); block 9’s raw winner (SBPLfree) rails a bound and is disclosed invalid (best valid: CPL+BB at $\Delta\text{AIC} = 2.0$).

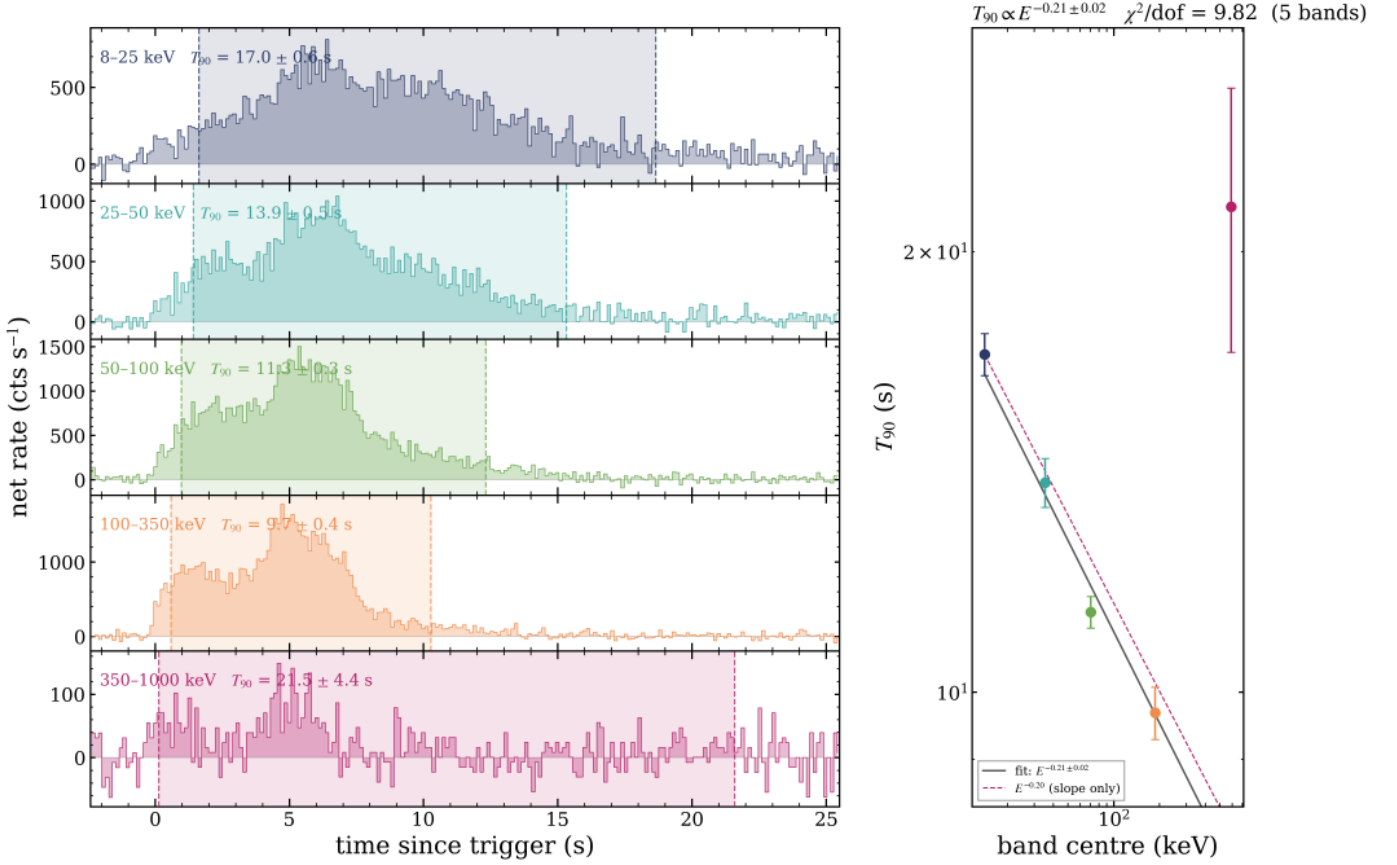
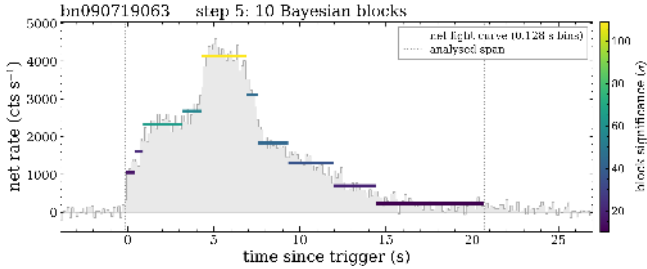
5.4. Thermal Candidates

Three blocks prefer BB-bearing models, every one a statistical tie: block 0 CPL+BB ($kT = 105 \pm 19$ keV, margin 1.6), block 4 SBPL+BB ($kT = 8.7 \pm 1.2$ keV, margin 0.1), block 6 Band+BB ($kT = 31.7 \pm 2.4$ keV, margin 0.8; Figure 11). All three Wien peaks sit clear of the 8 keV edge (411, 34, 124 keV), so none is an edge artifact — but none carries a decisive margin either, and the three temperatures do not form a coherent track (unlike burst #4’s monotonic cooling). The campaign accumulator records three edge-clean tie-level candidates and zero robust thermal detections for this burst. (The step-9 scorecard’s nested-LRT gate, a different metric, marks blocks 0, 2, and 4 — agreeing with the AIC-winner set only on blocks 0 and 4: one more instance of metric-dependence for the campaign synthesis.)

6. QUALITY CONTROL AND PROVENANCE

Step-9 QC (Figure 12) is built from the canonical campaign fit table; every figure carries its producing script’s SHA-256 in a JSON sidecar; numbers are recomputed from product files; all figures pass a fresh-context vision gate before delivery, enforced by a pre-delivery hook. Fits come from the campaign’s one-invocation 24-model engine run with best-minimum-per-(bin, model) canonicalization.

7. SUMMARY

bn090719063 — step 7: energy-resolved T_{90} (detector n8); dashed = t_5 , t_{95} **Figure 7.** Per-band light curves with individual T_{90} spans and the $T_{90}(E)$ relation.**Figure 8.** Bayesian-block spectral bins over the light curve.

1. GRB 090719 is a compact ($T_{90} = 13.39 \pm 0.22$ s) single pulse with the second consecutive *mixed* (non-FRED) shape ($\varphi = 0.408 \pm 0.023$) — two of five analysed pulses now cluster near $\varphi = 0.4$.
2. The canonical MVT is the campaign's fastest (8.9 ± 1.9 ms, detection, $z = 2.8$ caveat); the CWT lands on a new grid rung (256 ms after four bursts at 215 ms), mapping the estimator's discrete ladder; the intra-burst estimator spread reaches a factor of 76.
3. The lag is the campaign's most significant ($\tau = 0.80^{+0.03}_{-0.02} \pm 0.08_{\text{win}}$ s at 93σ); $\tau/T_{90} \approx 6.0\%$ extends the non-constant lag-width sequence.

4. Ten-block spectroscopy shows the cleanest hard-to-soft evolution of the campaign: E_p falls near-monotonically $624 \rightarrow \leq 30$ keV (factor ≥ 21 ; the endpoint fit is bound-railed) and α crosses the line of death mid-burst — the third consecutive epoch-level violation.
5. The integrated spectrum is a free-scale SBPL ($E_b = 188^{+30}_{-18}$ keV) in a $\Delta\text{AIC} = 0.2$ tie with a BB-bearing composite — reported unresolved.
6. Three BB-bearing block winners are all tie-level ($\Delta\text{AIC} \leq 1.6$), all edge-clean, forming no coherent temperature track: zero robust thermal detections.
7. The integrated winner is the campaign's first off-rail fit ($n_8 = 1.01$, $b_1 = 1.14$), though block-level winners still rail b_1 in six of ten blocks; block 9's bound-railed invalid winner and two labeled engine-refusal cells are disclosed.

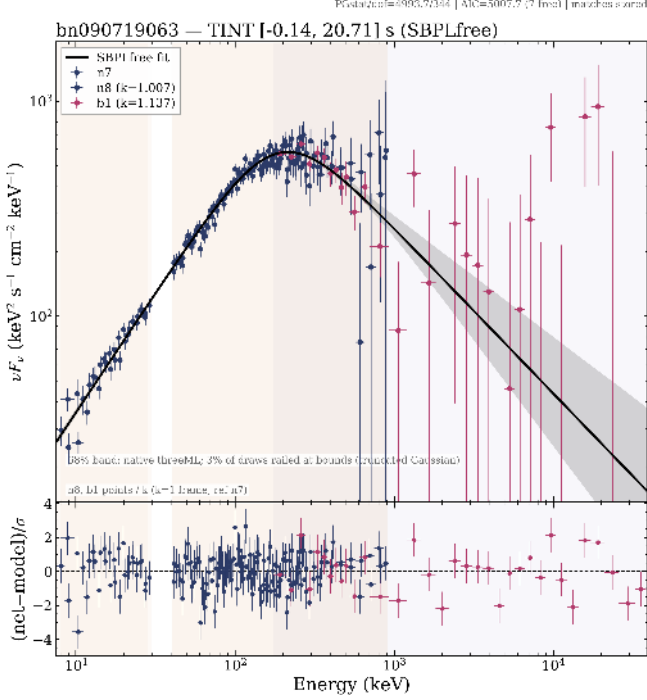


Figure 9. Integrated-spectrum winner: free-scale SBPL with its break at 188 keV; the BB-bearing runner-up sits at $\Delta\text{AIC} = 0.2$ (statistical tie). 68% band from propagated posterior draws; PGstat/dof, AIC, and EACs stamped on-figure.

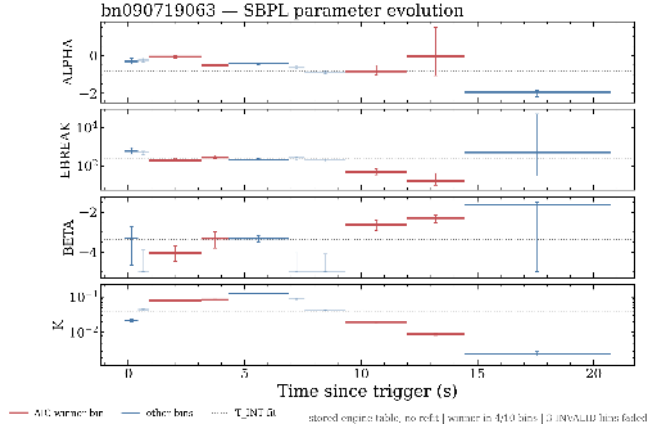


Figure 10. SBPL-family parameter evolution across the ten blocks: the E_b decay (block-3 excursion aside) and the α softening; the line-of-death crossing is quoted per block in the text.

APPENDIX

A. ALL-MODEL MONTAGES

Complete 24-model montages for the integrated spectrum and all ten blocks; the winner panel is marked in each; the two engine-refusal cells carry labeled refusal panels.

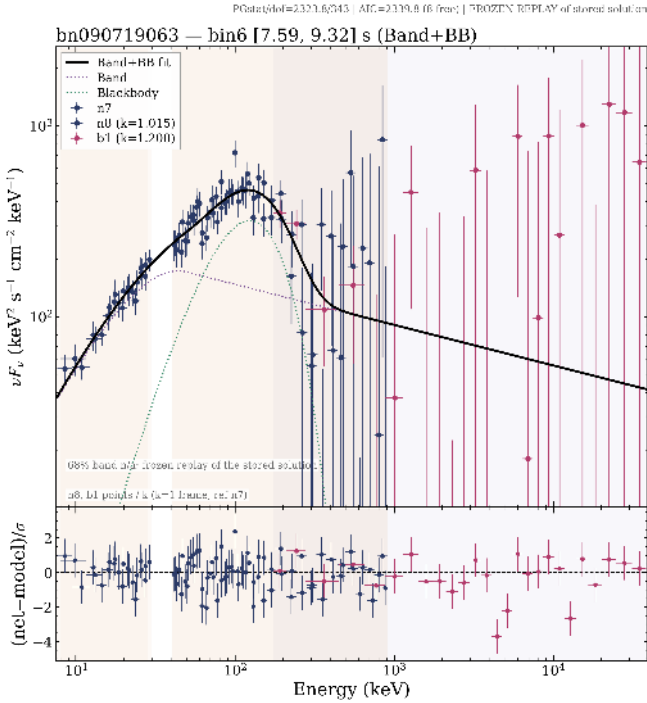


Figure 11. Block 6: Band+BB tie winner ($kT = 31.7$ keV, components dotted) — edge-clean but margin $\Delta\text{AIC} = 0.8$ only.

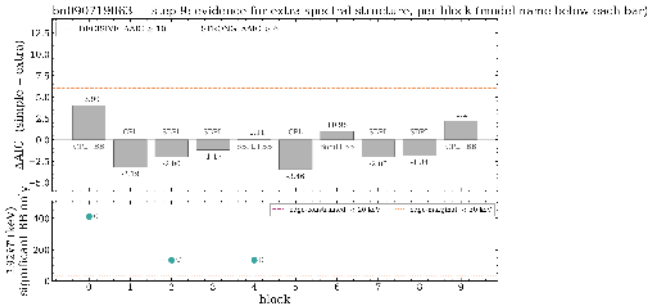


Figure 12. Step-9 scorecard: per-block evidence for extra spectral structure (top) and the $3.92 kT$ Wien-peak edge check (bottom).

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bn090719063 — TINT [-0.14, 20.71] s — 24 models, AIC-ordered — winner SBPLF

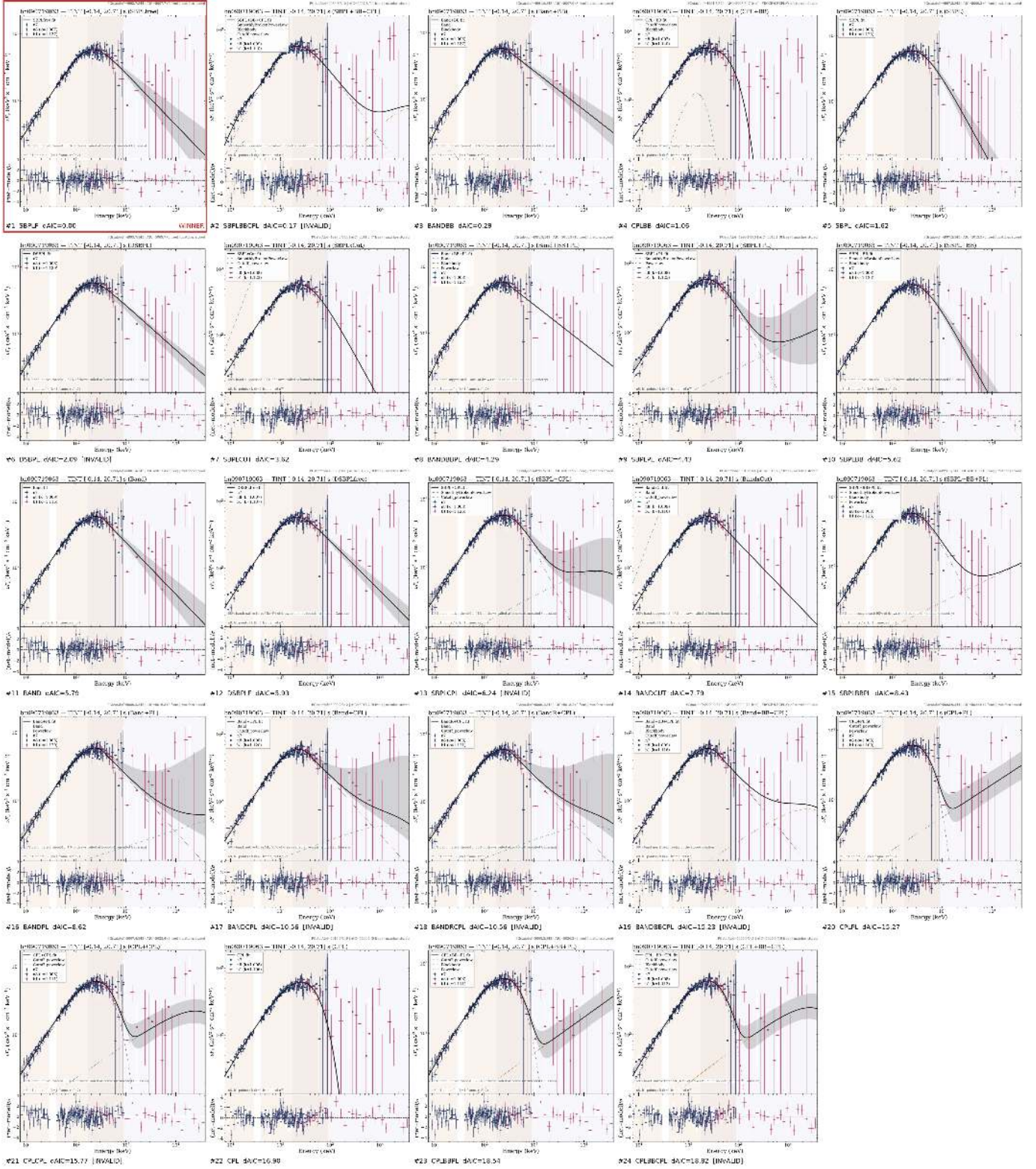


Figure 13. Integrated spectrum: all 24 models.

bn090719063 — bin0 [-0.14, 0.42] s — 24 models, AIC-ordered — winner CPLBB

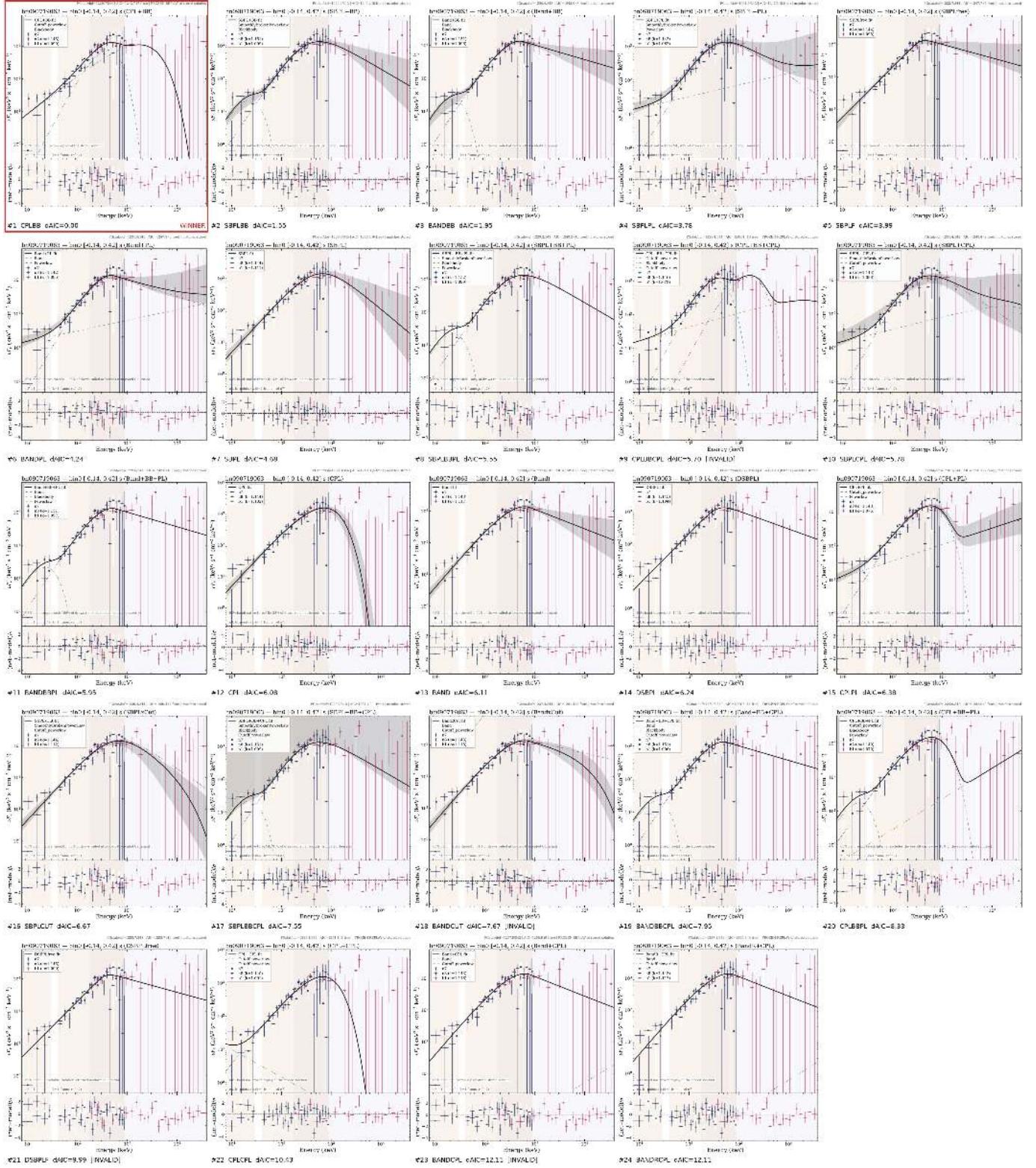


Figure 14. Block 0: all 24 models.

bn090719063 — bin1 [0.42, 0.89] s — 24 models, AIC-ordered — winner CPL

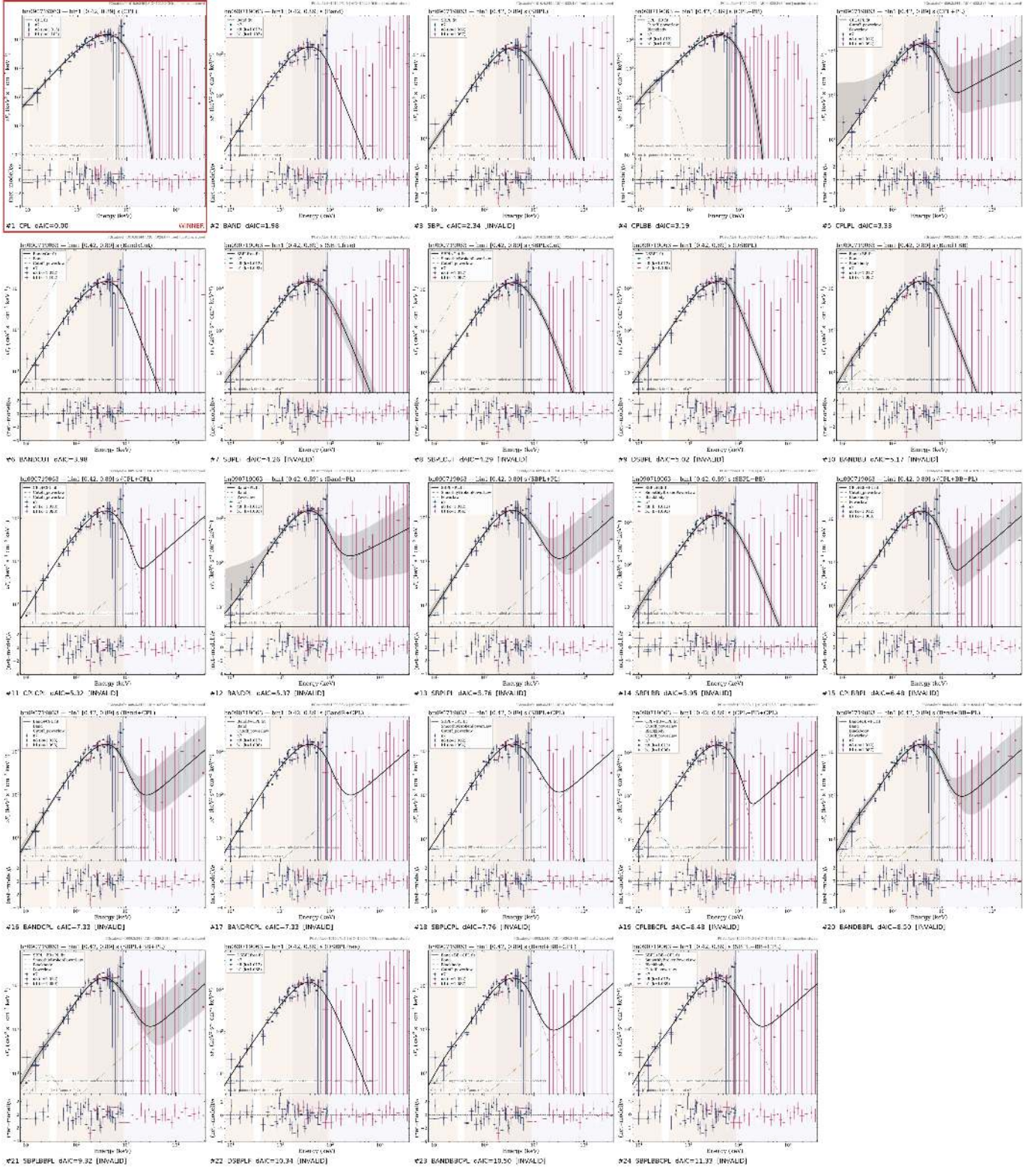


Figure 15. Block 1: all 24 models.

bn090719063 — bin2 [0.89, 3.16] s — 24 models, AIC-ordered — winner SBPL

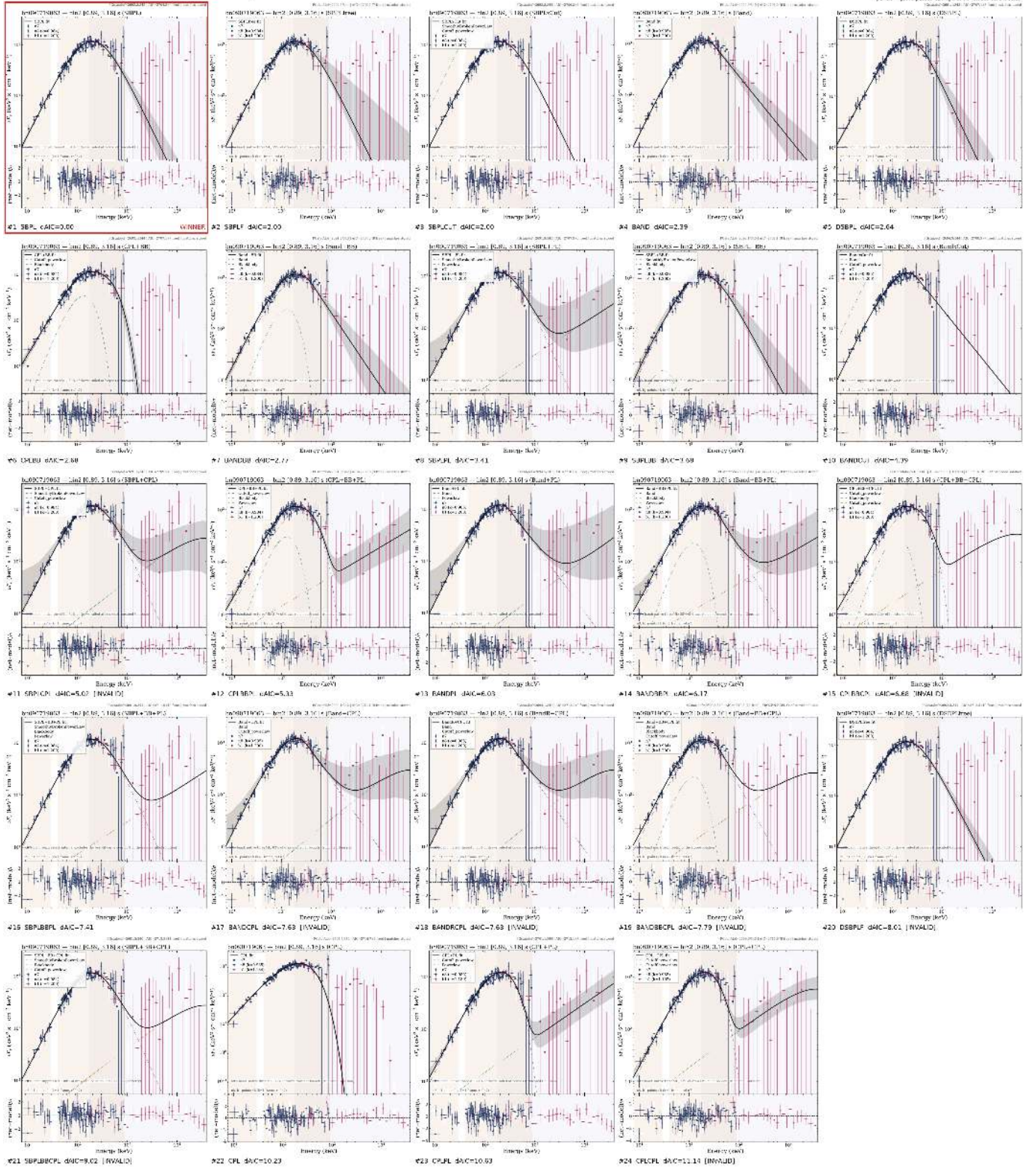


Figure 16. Block 2: all 24 models.

bn090719063 — bin3 [3.16, 4.30] s — 24 models, AIC-ordered — winner SBPL

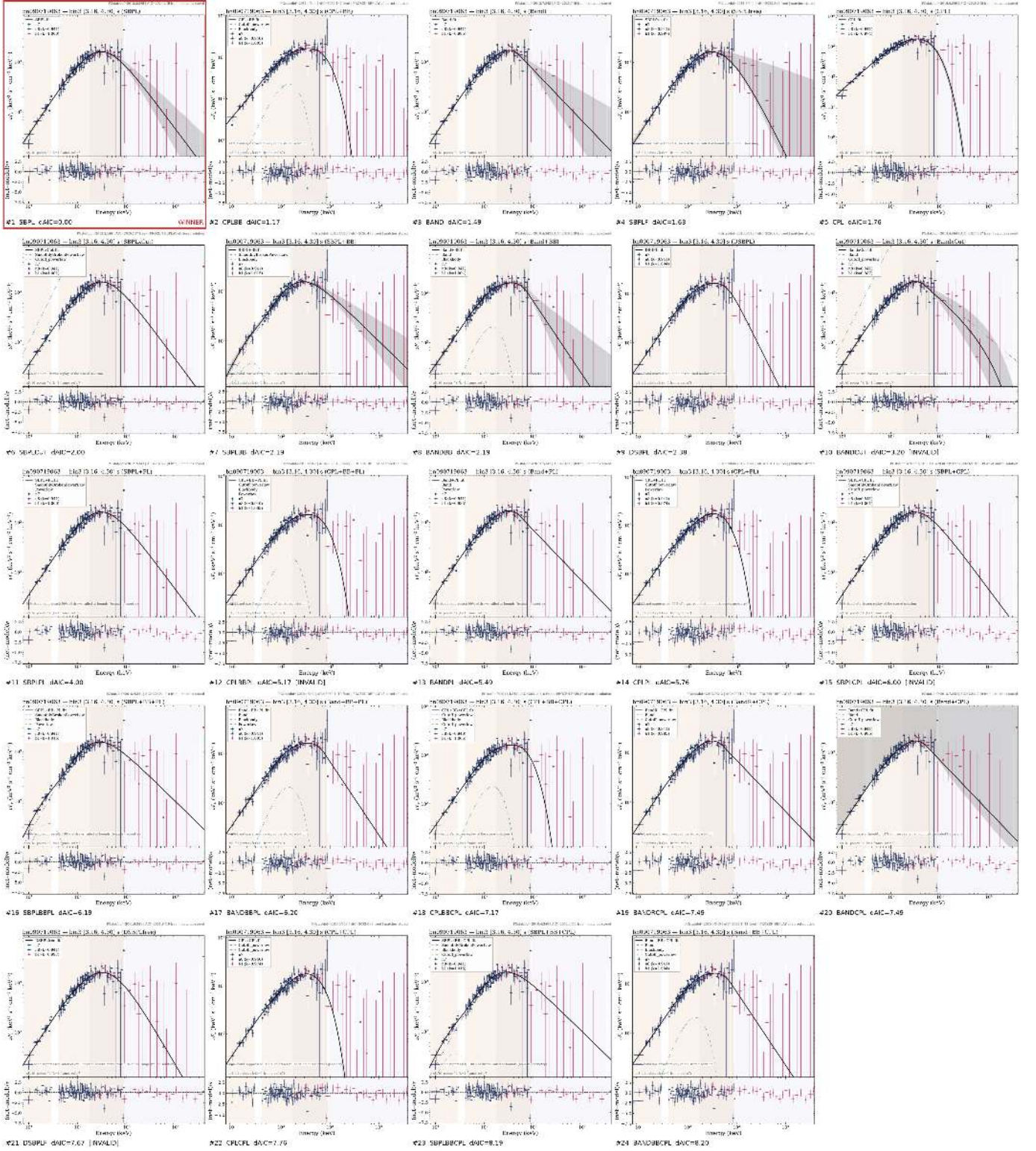


Figure 17. Block 3: all 24 models.

bn090719063 — bin4 [4.30, 6.87] s — 24 models, AIC-ordered — winner SBPLBB

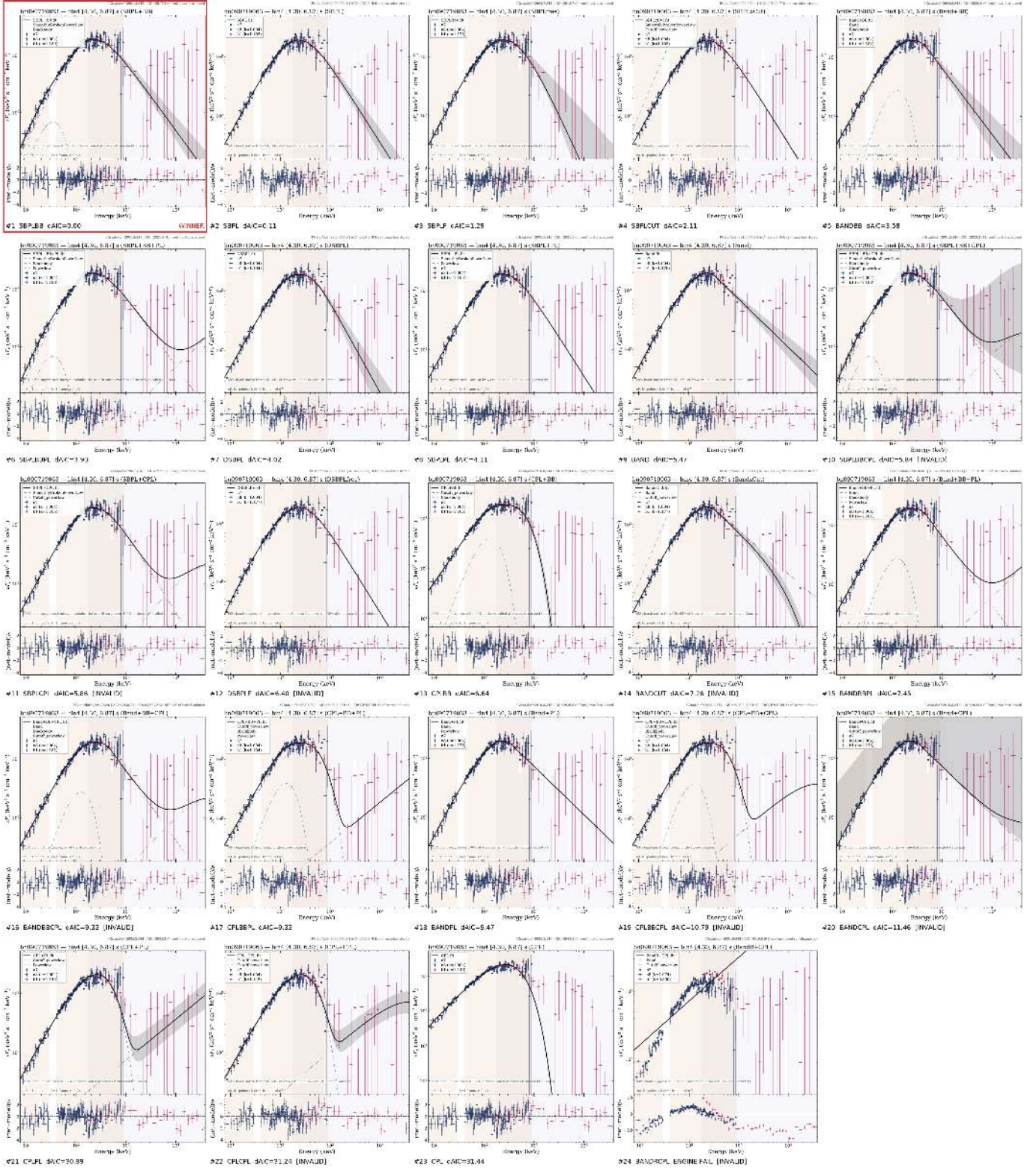


Figure 18. Block 4: all 24 models (BandR+CPL: labeled refusal cell).

bn090719063 — bin5 [6.87, 7.59] s — 24 models, AIC-ordered — winner CPL

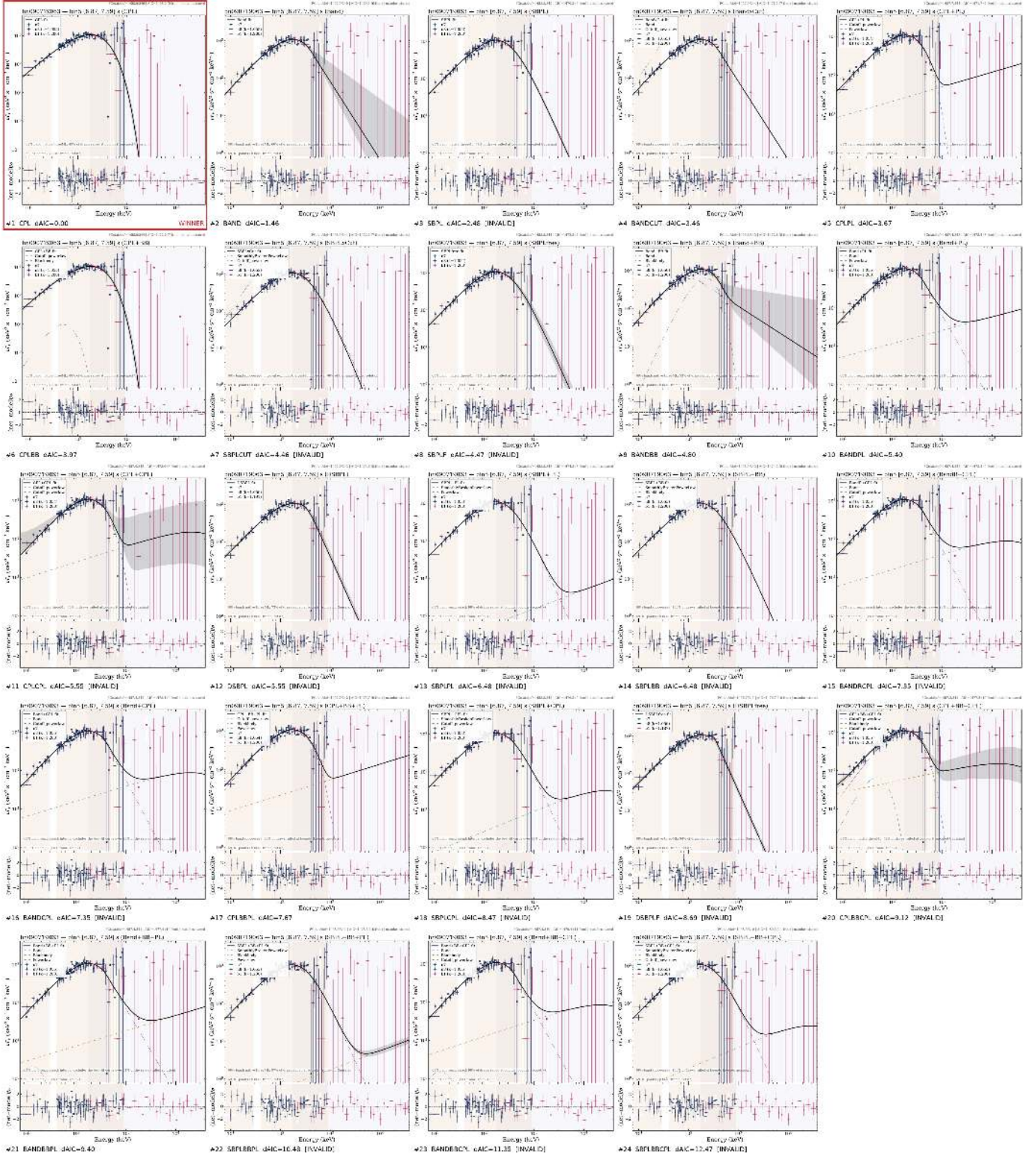


Figure 19. Block 5: all 24 models.

bn090719063 — bin6 [7.59, 9.32] s — 24 models, AIC-ordered — winner BANDBB



Figure 20. Block 6: all 24 models.

bn090719063 — bin7 [9.32, 11.98] s — 24 models, AIC-ordered — winner SBPL



Figure 21. Block 7: all 24 models.

bn090719063 — bin8 [11.98, 14.44] s — 24 models, AIC-ordered — winner SBPL

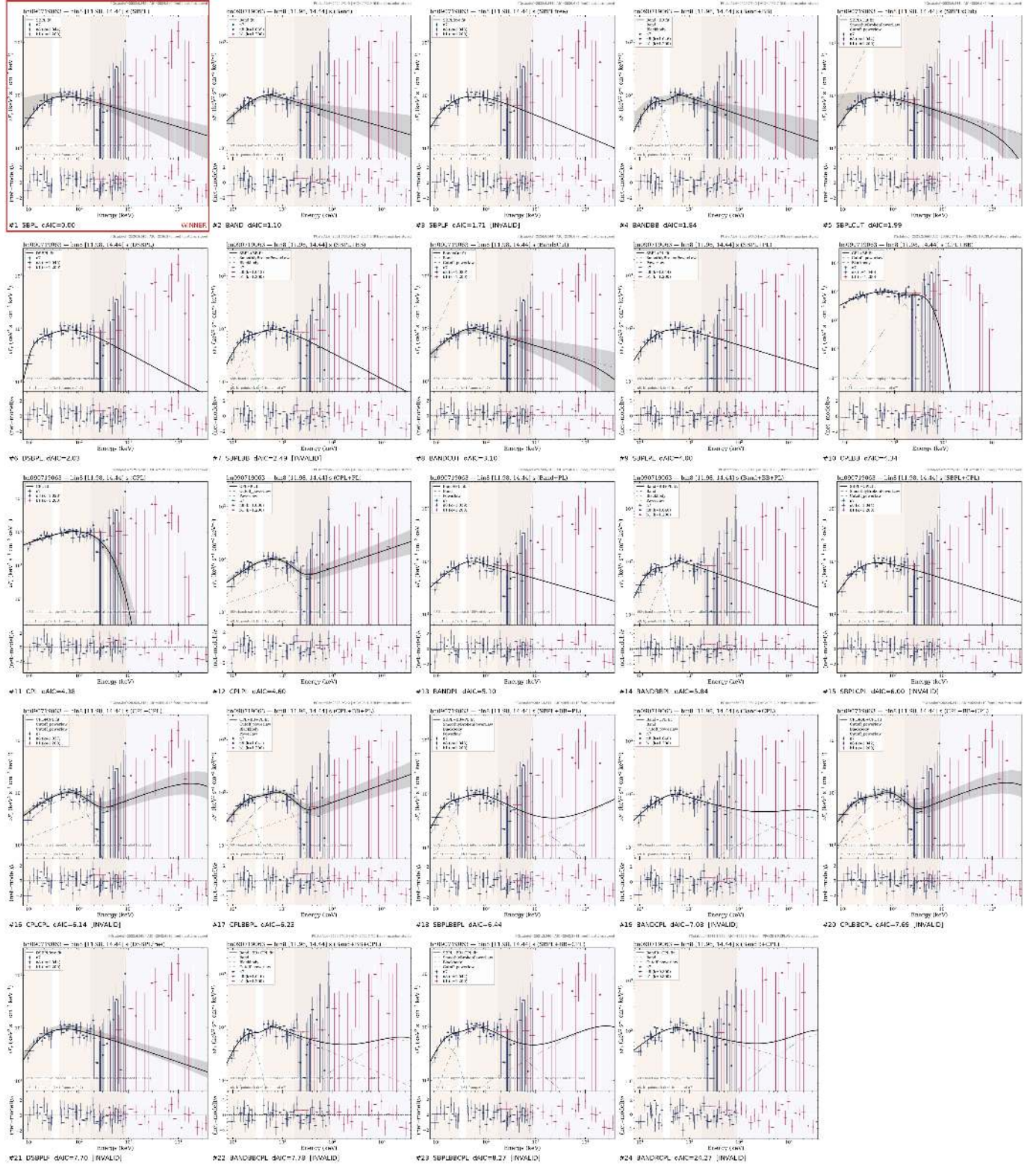


Figure 22. Block 8: all 24 models.

bn090719063 — bin9 [14.44, 20.71] s — 24 models, AIC-ordered — winner SBPLF

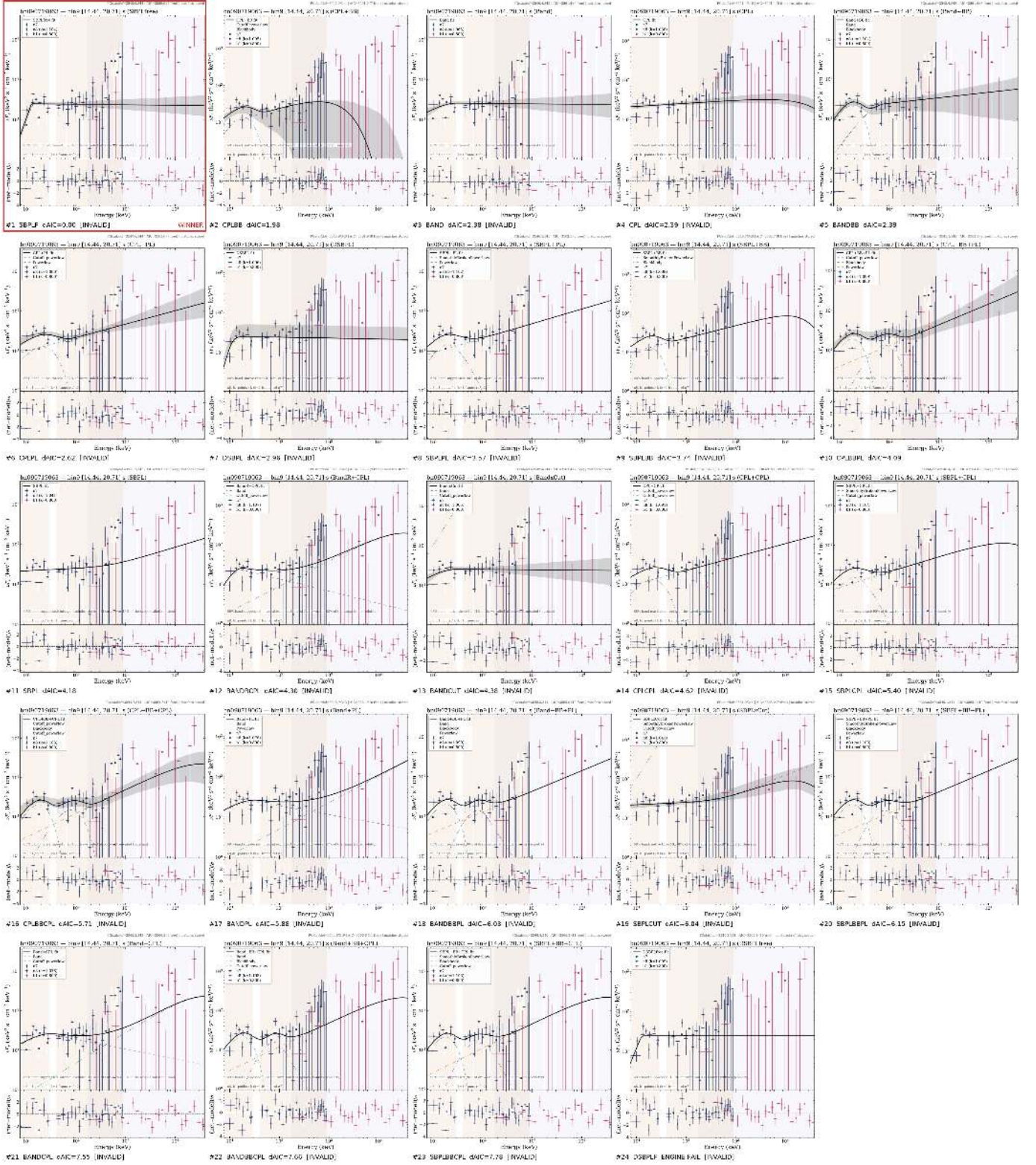


Figure 23. Block 9: all 24 models (DSBPLfree: labeled refusal cell).